PL/SQL PROGRAMMING

Exercise 1: Control Structures

Objective:

To implement PL/SQL control structures such as `IF-THEN-ELSE`, `CASE`, and looping constructs for decision-making and iteration.

Code:

```
DECLARE
    num NUMBER := 15;
BEGIN
    IF num < 10 THEN
        DBMS_OUTPUT.PUT_LINE('Number is less than 10');
    ELSIF num BETWEEN 10 AND 20 THEN
        DBMS_OUTPUT.PUT_LINE('Number is between 10 and 20');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Number is greater than 20');
    END IF;
END;
/
```

Program Output Screenshot:

Output Text:

```
Number is between 10 and 20
```

Result:

The program correctly executes conditional logic using PL/SQL control structures.

Conclusion:

Control structures enable dynamic decision-making and flow control in PL/SQL blocks.

Exercise 3: Stored Procedures

Objective:
To create and execute a stored procedure that calculates the total salary of employees from a database table.

Code:

```
CREATE OR REPLACE PROCEDURE total_salary IS
    total NUMBER;
BEGIN
    SELECT SUM(salary) INTO total FROM employees;
    DBMS_OUTPUT.PUT_LINE('Total Salary: ' || total);
END;
/
EXEC total_salary;
```

Output Text:

```
Total Salary: 450000
```

Result:
The stored procedure successfully computes and displays the total salary from the employee table.

Conclusion:
Stored procedures simplify repetitive tasks and improve database performance through modular execution.